Meeting 1 — Multivariable Functions, Partial Derivatives, and Extrema

Mathematics for Chemists | 2.5–3 hours

Part 1 — Theory and Algorithm (~45 min)

1.1 A Chemical Question First

Before any definitions, open with a single question:

"A reaction mixture is prepared. Where does it end up?"

The answer — in any undergraduate thermodynamics course — is: at the minimum of the Gibbs free energy G. The system "rolls downhill" on the G surface until it can go no lower. This is not a metaphor. The surface is a real function of real variables, and finding its minimum is a precise mathematical problem. Everything in today's meeting is a toolkit for solving that problem.

1.2 Warm-Up: The One-Variable Story

Recall the standard setup. For $f\colon \mathbb{R} \to \mathbb{R}$, we say x_0 is a **critical point** if $f'(x_0) = 0$. The second derivative then classifies it:

$$f''(x_0) > 0 \;\Rightarrow\; \text{local minimum}, \qquad f''(x_0) < 0 \;\Rightarrow\; \text{local maximum}, \qquad f''(x_0) = 0 \;\Rightarrow\; \text{inconclusive}.$$

Example: the Morse potential. The interaction energy between two atoms in a diatomic molecule is modelled as

$$U(r) = D_e\left(1 - e^{-a(r-r_e)}\right)^2,$$

where r is the internuclear distance, r_e the equilibrium bond length, D_e the dissociation energy, and a controls the well width. Computing $U'(r) = 0$ gives $r = r_e$; checking $U''(r_e) = 2D_e a^2 > 0$ confirms a minimum. The chemistry is encoded in the mathematics.

> *Sketch on board: the Morse potential curve; mark r_e, D_e, and the asymptote at $U = D_e$.*

Why one variable is not enough. A water molecule has three nuclei, hence $3 \times 3 = 9$ positional degrees of freedom (reduced to 3 internal after removing translations and rotations: two bond lengths and one angle). A benzene molecule has 12 internal degrees of freedom. In general, a molecule with N atoms has a potential energy surface in $3N - 6$ dimensions. We cannot reduce this to a single variable. We need a general theory.

1.3 Functions of Several Variables

Let $f\colon \mathbb{R}^n \to \mathbb{R}$, i.e., f takes a vector $(x_1, \ldots, x_n)$ and returns a single number. For $n = 2$ the graph $z = f(x, y)$ is a surface in $\mathbb{R}^3$.

Key geometric objects.

- **Graph:** $\{(x, y, f(x, y)) : (x, y) \in \mathbb{R}^2\}$ — a surface.
- **Level curves (contour lines):** $\{(x, y) : f(x, y) = c\}$ for various constants c. These are the lines on a topographic map. Chemists encounter them constantly: isotherms, isobars, and phase boundaries on a phase diagram are all level curves of thermodynamic functions.

> *Sketch on board: a surface with a bowl shape, and below it the corresponding circular contour lines closing around the minimum.*

Partial derivatives. The partial derivative with respect to x_i is the ordinary derivative obtained by treating all other variables as constants:

$$\frac{\partial f}{\partial x_i}(x_0) = \lim_{h \to 0} \frac{f(x_0 + h\,\mathbf{e}_i) - f(x_0)}{h},$$

where $\mathbf{e}_i$ is the i-th standard basis vector. Geometrically: slice the surface with the plane parallel to the x_i-axis through x_0, and take the slope of the resulting curve.

The gradient. Collect all partial derivatives into a vector:

$$\nabla f = \left(\frac{\partial f}{\partial x_1}, \ldots, \frac{\partial f}{\partial x_n} \right).$$

The gradient points in the direction of **steepest ascent**; its magnitude $\|\nabla f\|$ is the rate of increase in that direction. Consequently, $-\nabla f$ is the direction of steepest descent — the direction a ball would roll, or the direction a reaction mixture moves on a free energy surface to lower G.

Critical points. A point x_0 is a **critical point** of f if $\nabla f(x_0) = \mathbf{0}$, i.e., all partial derivatives vanish simultaneously. These are the candidates for local extrema and saddle points.

1.4 The Hessian Matrix and the Second Derivative Test

At a critical point, the gradient gives no information about the nature of the point. We need the second-order data. Define the **Hessian matrix**:

$$H(f)(x_0) = \left. \begin{pmatrix} \dfrac{\partial^2 f}{\partial x_1^2} & \dfrac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots \\ \dfrac{\partial^2 f}{\partial x_2 \partial x_1} & \dfrac{\partial^2 f}{\partial x_2^2} & \cdots \\ \vdots & & \ddots \end{pmatrix} \right|_{x_0}.$$

By Schwarz's theorem (smoothness of f), H is **symmetric**: $\partial^2 f/\partial x_i \partial x_j = \partial^2 f/\partial x_j \partial x_i$. Hence H has real eigenvalues.

The second derivative test (general statement). At a critical point x_0, the local behaviour of f is governed by the signs of the eigenvalues $\lambda_1, \ldots, \lambda_n$ of $H(x_0)$:

Eigenvalues	Type
All $\lambda_i > 0$	**Local minimum** (H positive definite)
All $\lambda_i < 0$	**Local maximum** (H negative definite)
Mixed signs	**Saddle point** (H indefinite)
Any $\lambda_i = 0$	**Inconclusive** — test fails

For $n = 2$ the eigenvalue conditions translate to the determinant and trace of H, which are more practical to compute:

$$D = \det H = \frac{\partial^2 f}{\partial x^2}\frac{\partial^2 f}{\partial y^2} - \left(\frac{\partial^2 f}{\partial x \partial y}\right)^2, \qquad T = \operatorname{tr} H = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}.$$

D	T	Conclusion
$D > 0$, $T > 0$		Local minimum
$D > 0$, $T < 0$		Local maximum
$D < 0$		Saddle point
$D = 0$		Inconclusive

(When $D > 0$ both eigenvalues share a sign, and T determines which.)

1.5 Connection to ODE Stability — The Same Mathematical Structure

> *Conceptual bridge — spend about 5 minutes here.*

Students who have studied $\dot{\mathbf{x}} = A\mathbf{x}$ have already classified fixed points via eigenvalues of A: stable node ($\lambda_i < 0$), unstable node ($\lambda_i > 0$), saddle (mixed), centre (purely imaginary). The Hessian test is **structurally identical**.

This is not a coincidence. Near a critical point x_0, any smooth function is well-approximated by its quadratic Taylor expansion:

$$f(x_0 + \mathbf{h}) \approx f(x_0) + \underbrace{\nabla f(x_0)}_{=0} \cdot \mathbf{h} + \tfrac{1}{2}\,\mathbf{h}^T H(x_0)\,\mathbf{h}.$$

The behaviour of $f - f(x_0)$ near x_0 is entirely determined by the quadratic form $Q(\mathbf{h}) = \mathbf{h}^T H\mathbf{h}$. Classifying this form by the signs of eigenvalues of H is precisely **Sylvester's criterion for quadratic forms** — and it is the same spectral analysis that classifies fixed points of linear ODE systems.

The unifying principle: **stability, in mathematics, is always a question about the sign of a symmetric bilinear form.** Whether we ask "does this reaction reach equilibrium?", "is this fixed point stable?", or "is this critical point a minimum?" — we are always asking whether a certain symmetric matrix is positive definite.

1.6 The Algorithm for Finding Local Extrema

Algorithm. Let $f\colon \mathbb{R}^n \to \mathbb{R}$ be twice continuously differentiable.

Step 1. Compute all partial derivatives $\dfrac{\partial f}{\partial x_1}, \dots, \dfrac{\partial f}{\partial x_n}$.

Step 2. Solve the system $\nabla f = \mathbf{0}$, i.e. $\dfrac{\partial f}{\partial x_1} = 0, \ \dots, \ \dfrac{\partial f}{\partial x_n} = 0$. Each solution is a critical point candidate.

Step 3. For each critical point x_0, compute $H(x_0)$.

Step 4. Classify $H(x_0)$:

- For $n = 2$: compute $D = \det H$ and $T = \operatorname{tr} H$; apply the table above.
- For general n: compute eigenvalues of H (or use Sylvester's criterion on

leading principal minors).

Step 5. If $\det H(x_0) = 0$: **the test is inconclusive.** Proceed to higher-order analysis or geometric reasoning (see §1.7).

Example 1. $f(x, y) = x^2 + y^2 - 2x + 4y + 5$.

Step 1. $\partial f/\partial x = 2x - 2, \quad \partial f/\partial y = 2y + 4$.

Step 2. $x = 1$, $y = -2$. One critical point: $(1, -2)$.

Step 3. $H = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$ (constant).

Step 4. $D = 4 > 0$, $T = 4 > 0$: **local minimum**. $f(1, -2) = 0$.

Remark: completing the square gives $f = (x-1)^2 + (y+2)^2 \geq 0$, confirming this is a global minimum.

Example 2. $f(x, y) = x^2 - y^2$.

$\nabla f = (2x, -2y) = \mathbf{0} \Rightarrow (0, 0)$. $\quad H = \begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix}$. $\quad D = -4 < 0$: **saddle point**. (A mountain pass: minimum along x, maximum along y.)

1.7 When the Test Fails: $\det H = 0$

A vanishing determinant means at least one eigenvalue is zero: the quadratic approximation is flat in that direction, and higher-order terms determine the behaviour. The test gives **no information** — all three outcomes (minimum, maximum, saddle) are possible.

Example 1 — a minimum. $f(x, y) = x^4 + y^4$.

$\nabla f = (4x^3, 4y^3) = \mathbf{0}$ only at $(0, 0)$. $H\big|_{(0,0)} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$, $\det H = 0$. Yet $f(x, y) \geq 0 = f(0, 0)$ everywhere: **global minimum**.

Example 2 — a saddle. $g(x, y) = x^3 - 3xy^2$ (the "monkey saddle").

$\nabla g = \mathbf{0}$ at $(0, 0)$; $H\big|_{(0,0)} = 0$. But $g(r, 0) = r^3 > 0$ while $g(r, r) = -2r^3 < 0$ for $r > 0$: the function changes sign in every neighbourhood of the origin. **Saddle.**

Example 3 — another saddle. $h(x, y) = x^2 y$.

$\nabla h = (2xy, x^2) = \mathbf{0}$ on the entire y-axis; $\det H = 0$. $h(x, \varepsilon) = \varepsilon x^2 \geq 0$ while $h(x, -\varepsilon) \leq 0$: the function takes both signs near $(0, 0)$. **Saddle.**

Conclusion. When $\det H = 0$, there is no shortcut. Options: (a) factor the function and inspect the sign; (b) restrict to lines through the critical point; (c) use a higher-order Taylor expansion. In practice one often argues from the geometry or symmetry of the problem.

Part 2 — Exercises (~60 min)

Find and classify all critical points of the following functions. For each, state whether the Hessian test is conclusive.

Exercise 1. $f(x, y) = x^2 + y^2 - 2x + 4y + 5$

Hint: one global minimum. Answer: $(1, -2)$, $f = 0$.

Exercise 2. $f(x, y) = x^2 - y^2$

Pure saddle at origin. Verify with both the det *test and by direct inspection of level curves.*

Exercise 3. $f(x, y) = x^3 - 3xy^2$

Monkey saddle at origin. $\det H = 0$*: the standard test fails. Use restriction to lines through the origin. Note that* $f = 0$ *on the three lines* $y = 0$, $y = \sqrt{3}\,x$, $y = -\sqrt{3}\,x$ *— these are the "three valleys" of the saddle.*

Exercise 4. $f(x, y) = e^{-(x^2+y^2)}$

In polar coordinates f *depends only on* $r = \sqrt{x^2 + y^2}$*. Global maximum at origin;* $f \to 0$ *at infinity. Challenge: what are the level curves? What does* ∇f *look like geometrically?*

Exercise 5. $f(x, y) = \sin x \cdot \sin y, \quad (x, y) \in [0, 2\pi]^2$

Critical points where $\cos x \sin y = 0$ *and* $\sin x \cos y = 0$*. Classify the four interior critical points. The pattern reveals the "checkerboard" structure of* $\sin x \sin y$*.*

Exercise 6. $f(x, y, z) = x^2 + 2y^2 + 3z^2 - xy + yz$

Three variables. $\nabla f = (2x - y,\ 4y - x + z,\ 6z + y) = \mathbf{0}$ *gives* $x = y = z = 0$*. Hessian:* $H = \begin{pmatrix} 2 & -1 & 0 \\ -1 & 4 & 1 \\ 0 & 1 & 6 \end{pmatrix}$*. Check leading principal minors:* $\Delta_1 = 2 > 0$, $\Delta_2 = 7 > 0$, $\Delta_3 = \det H$*.*
If all are positive: global minimum.

Exercise 7 (Chemistry). The simplified Gibbs free energy of mixing for a two-component system is

$$G(x, y) = x^2 + y^2 + xy - 3x - 3y,$$

where x, y are deviations of mole fractions from reference values. Find the equilibrium composition (minimum of G) and classify it. What would a saddle point of G represent physically?

Answer: $x = y = 1$, $H = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, $\det H = 3 > 0$, $\operatorname{tr} H = 4 > 0$*: minimum. A saddle would correspond to an unstable equilibrium: composition minimises* G *in one direction but not another.*

Exercise 8 (Implicit curves preview). Consider the curve $F(x, y) = x^2 + y^2 - 1 = 0$ (the unit circle).

(a) Find all points where $\partial F/\partial y = 0$. What happens to y as a function of x at these points?
(b) At all other points compute dy/dx by implicit differentiation of $F(x, y(x)) = 0$.
(c) Verify your formula at the point $(0, 1)$ by direct computation.

This previews the implicit function theorem: the condition $\partial F/\partial y \neq 0$ *is exactly what guarantees that* y *can locally be expressed as a smooth function of* x *along* $F = 0$*.*

Exercise 9 (Degenerate Hessian). $f(x, y) = x^4 + y^4 - 4xy$

$\nabla f = (4x^3 - 4y,\ 4y^3 - 4x) = \mathbf{0}$ *gives three critical points:* $(0, 0)$, $(1, 1)$, $(-1, -1)$*. At* $(0, 0)$*:* $H = \begin{pmatrix} 0 & -4 \\ -4 & 0 \end{pmatrix}$, $\det H = -16 < 0$*: saddle. At* $(\pm 1, \pm 1)$*:* $H = \begin{pmatrix} 12 & -4 \\ -4 & 12 \end{pmatrix}$, $\det H = 128 > 0$, $\operatorname{tr} H = 24 > 0$*: minima.*

Exercise 10 (Geometric reasoning). For $f(x, y) = x^2 e^{-(x^2+y^2)}$:

(a) Without computing, argue whether f has a global maximum and a global minimum.
(b) Sketch the level curves; identify regions where f increases or decreases moving away from the origin.
(c) Only then compute $\nabla f = \mathbf{0}$ and confirm your prediction.

Geometric argument: $f \geq 0$ everywhere and $f \to 0$ as $\|(x,y)\| \to \infty$, and $f = 0$ on the y-axis. A maximum must be achieved somewhere. By symmetry the maxima lie on the x-axis. Setting $\partial f/\partial x = 0$: $(2x - 2x^3)e^{-(x^2+y^2)} = 0 \Rightarrow x = \pm 1$, $y = 0$.

Part 3 — Advanced Applications in Chemistry (~30 min)

3.1 Potential Energy Surfaces and Transition State Theory (~12 min)

The **potential energy surface** (PES) of a molecule is the function $U\colon \mathbb{R}^{3N-6} \to \mathbb{R}$ mapping internal coordinates to total electronic energy. Finding the mechanism of a reaction means navigating this surface.

Stationary points and their classification.

Type	$\nabla U = 0$?	Hessian	Physical meaning
Minimum	Yes	Positive definite	Reactant, product, intermediate
Index-1 saddle	Yes	One $\lambda < 0$, rest > 0	Transition state
Index-k saddle	Yes	k negative eigenvalues	Higher-order saddle (rare)

The gradient $-\nabla U$ gives the forces on the nuclei (Newton's second law for nuclear motion under the Born–Oppenheimer approximation). A transition state is a saddle of index 1: minimum in all directions except one, the **reaction coordinate**. The single negative eigenvalue corresponds to the imaginary vibrational frequency $\nu^\dagger$ in transition state theory:

$$k = \frac{k_B T}{h} \cdot \frac{Z^\dagger}{Z_A Z_B} \cdot e^{-E_a/k_B T}.$$

The Hessian at a minimum, when divided by atomic masses (the **mass-weighted Hessian**), gives the force constant matrix. Its eigenvalues are the squared normal mode frequencies ω_i^2. This is why quantum chemistry packages compute the Hessian at optimised geometries: to confirm it is a minimum (all $\omega_i^2 > 0$) and to extract infrared and Raman spectra directly.

> *Key message: the spectral decomposition of the Hessian* is *experimental spectroscopy.*

3.2 Thermodynamic Potentials, Maxwell Relations, and $C_p - C_V$

This subsection contains more material than can be covered in 13 minutes; treat it as lecture notes to be drawn from selectively, with the $C_p - C_V$ derivation as the main payoff.

What does dH mean? Differential forms

H is a smooth function $H(S, p)\colon \mathbb{R}^2 \to \mathbb{R}$ — a function exactly in the sense of this course. Its **total differential**

$$dH = \frac{\partial H}{\partial S}\, dS + \frac{\partial H}{\partial p}\, dp$$

is a **linear form** (a 1-form): given a small displacement (dS, dp) in the (S,p)-plane, it returns the best linear approximation to the change in H. The symbols dS and dp are not "infinitely small numbers"; they are formal placeholders for the two independent displacement directions.

Writing $dH = T\,dS + V\,dp$ is therefore shorthand for two partial derivative identities:

$$\left.\frac{\partial H}{\partial S}\right|_p = T, \qquad \left.\frac{\partial H}{\partial p}\right|_S = V.$$

Exact vs. inexact differentials. Not every expression $M\,dx + N\,dy$ equals df for some function f. In thermodynamics, heat and work are written with δ rather than d:

$$\delta Q = T\,dS, \qquad \delta W = p\,dV.$$

These are **inexact**: there is no function "total heat" $Q(S,V)$ whose differential equals $T\,dS$. The heat absorbed along a process depends on the path, not just the endpoints. By contrast, dU, dH, dF, dG are **exact**: U, H, F, G are genuine functions of state, and their integrals along any path depend only on the endpoints.

The mathematical test for exactness is Schwarz's theorem: $M\,dx + N\,dy = df$ for some f if and only if $\partial M/\partial y = \partial N/\partial x$. This is precisely the condition from which the Maxwell relations follow.

The four thermodynamic potentials

Starting point: the combined first and second laws. For a reversible process in a closed system doing only pV-work, the second law gives $\delta Q_{\text{rev}} = T\,dS$ and the first law $dU = \delta Q - p\,dV$, so:

$$dU = T\,dS - p\,dV.$$

The natural variables of U are (S,V), and reading off partial derivatives: $(\partial U/\partial S)_V = T$, $(\partial U/\partial V)_S = -p$.

The remaining potentials are obtained by **Legendre transforms** — the same operation that converts the Lagrangian to the Hamiltonian in mechanics. Each transform replaces one natural variable with its conjugate:

Enthalpy $H = U + pV$ (replace $V \to p$):

$$dH = dU + p\,dV + V\,dp = T\,dS + V\,dp.$$

Helmholtz free energy $F = U - TS$ (replace $S \to T$):

$$dF = dU - T\,dS - S\,dT = -S\,dT - p\,dV.$$

Gibbs free energy $G = U - TS + pV$ (replace both):

$$dG = dU - T\,dS - S\,dT + p\,dV + V\,dp = -S\,dT + V\,dp.$$

Potential	Definition	Fundamental relation	Natural variables	Partial derivatives
U	—	$dU = T\,dS - p\,dV$	S, V	$T = \left(\dfrac{\partial U}{\partial S}\right)_V$, $-p = \left(\dfrac{\partial U}{\partial V}\right)_S$
H	$U + pV$	$dH = T\,dS + V\,dp$	S, p	$T = \left(\dfrac{\partial H}{\partial S}\right)_p$, $V = \left(\dfrac{\partial H}{\partial p}\right)_S$
F	$U - TS$	$dF = -S\,dT - p\,dV$	T, V	$-S = \left(\dfrac{\partial F}{\partial T}\right)_V$, $-p = \left(\dfrac{\partial F}{\partial V}\right)_T$
G	$U - TS + pV$	$dG = -S\,dT + V\,dp$	T, p	$-S = \left(\dfrac{\partial G}{\partial T}\right)_p$, $V = \left(\dfrac{\partial G}{\partial p}\right)_T$

Maxwell relations from Schwarz's theorem

Each fundamental relation $d\phi = M\,dx + N\,dy$ is exact, so $\partial M/\partial y = \partial N/\partial x$:

Potential	From	Maxwell relation
$U(S,V)$	$T\,dS - p\,dV$	$\left(\dfrac{\partial T}{\partial V}\right)_S = -\left(\dfrac{\partial p}{\partial S}\right)_V$
$H(S,p)$	$T\,dS + V\,dp$	$\left(\dfrac{\partial T}{\partial p}\right)_S = \left(\dfrac{\partial V}{\partial S}\right)_p$
$F(T,V)$	$-S\,dT - p\,dV$	$\left(\dfrac{\partial S}{\partial V}\right)_T = \left(\dfrac{\partial p}{\partial T}\right)_V$
$G(T,p)$	$-S\,dT + V\,dp$	$\left(\dfrac{\partial S}{\partial p}\right)_T = -\left(\dfrac{\partial V}{\partial T}\right)_p$

Every one of these is the statement $\partial^2\phi/\partial x_1\partial x_2 = \partial^2\phi/\partial x_2\partial x_1$ for the appropriate ϕ. The Maxwell relations are not special identities — they are the proof that the potentials are genuine functions of state.

> *Pause here and ask: "Have you used Maxwell relations in thermodynamics?" They have. "Did anyone explain why they hold?" Likely no. This is why mathematics matters.*

Derivation of $C_p - C_V$

This is the main payoff: a formula that cannot be derived without Maxwell relations, yet is standard in every physical chemistry course.

Step 1 — Express $C_p - C_V$ in terms of entropy. From $\delta Q_{\text{rev}} = T\,dS$ and the definition $C = \delta Q/dT$:

$$C_V = T\left(\frac{\partial S}{\partial T}\right)_V, \qquad C_p = T\left(\frac{\partial S}{\partial T}\right)_p, \qquad \text{so} \qquad C_p - C_V = T\left[\left(\frac{\partial S}{\partial T}\right)_p - \left(\frac{\partial S}{\partial T}\right)_V\right].$$

Step 2 — Relate the two entropy derivatives by the chain rule. Treat S as a function of T and V:

$$dS = \left(\frac{\partial S}{\partial T}\right)_V dT + \left(\frac{\partial S}{\partial V}\right)_T dV.$$

Divide by dT at constant p:

$$\left(\frac{\partial S}{\partial T}\right)_p = \left(\frac{\partial S}{\partial T}\right)_V + \left(\frac{\partial S}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_p.$$

Substituting:

$$C_p - C_V = T\left(\frac{\partial S}{\partial V}\right)_T \left(\frac{\partial V}{\partial T}\right)_p.$$

Step 3 — Apply the Maxwell relation from F. From the table: $\left(\frac{\partial S}{\partial V}\right)_T = \left(\frac{\partial p}{\partial T}\right)_V$. Therefore:

$$C_p - C_V = T\left(\frac{\partial p}{\partial T}\right)_V \left(\frac{\partial V}{\partial T}\right)_p.$$

Step 4 — Rewrite in terms of directly measurable quantities. Define the **thermal expansion coefficient** $\alpha = \frac{1}{V}\left(\frac{\partial V}{\partial T}\right)_p$ and the **isothermal compressibility** $\kappa_T = -\frac{1}{V}\left(\frac{\partial V}{\partial p}\right)_T$. Apply the cyclic identity (which holds whenever p, V, T are linked by an equation of state):

$$\left(\frac{\partial p}{\partial T}\right)_V \left(\frac{\partial T}{\partial V}\right)_p \left(\frac{\partial V}{\partial p}\right)_T = -1, \quad \text{so} \quad \left(\frac{\partial p}{\partial T}\right)_V = -\frac{(\partial V/\partial T)_p}{(\partial V/\partial p)_T} = \frac{\alpha V}{\kappa_T}.$$

Substituting:

$$\boxed{C_p - C_V = \frac{TV\alpha^2}{\kappa_T}.}$$

Consequences. Since $\kappa_T > 0$ (mechanically stable systems compress under pressure), $C_p \geq C_V$ always, with equality only if $\alpha = 0$ (no thermal expansion — water near 4 °C). For an ideal gas: $pV = nRT$ gives $\alpha = 1/T$ and $\kappa_T = 1/p$, so $C_p - C_V = nR$ — the familiar result recovered as a special case, with no additional assumptions.

The entire derivation used only three ingredients: the definition of dF, one Maxwell relation, and the chain rule. No model, no approximation.

3.3 Spinodal Decomposition (~5 min, closing remark)

For a binary mixture at fixed T, the Helmholtz free energy of mixing $\Delta F_{\text{mix}}(x)$ as a function of composition $x \in [0, 1]$ determines miscibility. The mixture is **locally stable** at composition x if and only if

$$\frac{\partial^2 \Delta F_{\text{mix}}}{\partial x^2} > 0.$$

This is the positive-definiteness condition for a 1×1 Hessian. When this second derivative is negative, even a small fluctuation in composition grows spontaneously: the mixture decomposes into two phases without any energy barrier. The curve in the (T, x) plane where $\partial^2 F/\partial x^2 = 0$ is called the **spinodal** — a level curve of the second derivative, separating stable from unstable compositions.

The same mathematical condition that tells us whether a critical point is a minimum or a saddle determines whether a material is stable or will spontaneously phase-separate.

Summary and Bridge to Meeting 2

What we did today. We defined critical points via $\nabla f = \mathbf{0}$, classified them using the spectral structure of the Hessian, and saw that this is the same quadratic-form analysis underlying

ODE stability theory. We applied the ideas to transition state theory, thermodynamic identities, and phase stability.

The algorithm (commit to memory)

1. Compute ∇f.
2. Solve $\nabla f = \mathbf{0}$ — find all critical points.
3. Compute H at each critical point.
4. Classify by eigenvalues (or $\det H$, $\operatorname{tr} H$ for $n = 2$).
5. If $\det H = 0$: the test fails, proceed with care.

Looking ahead. The level curves $f(x, y) = c$ we drew today are implicit relations. When is such a curve actually a smooth curve? When can we locally write y as a function of x along it? The answer — the **implicit function theorem** — requires $\nabla f \neq \mathbf{0}$ at the point in question. We will also meet different coordinate systems (polar, cylindrical, spherical), which are essential for functions with rotational symmetry — exactly the Morse potential, the hydrogen atom, and the spherical harmonics.